

INTRODUCTION TO LARGE LANGUAGE MODELS

Generative AI for Text

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Introduction to Large Language Models

Generative AI for Text

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Endorsement

‘Introduction to Large Language Models’ by Tanmoy Chakraborty is the go-to textbook for students, researchers, and professionals eager to delve into the rapidly evolving world of AI-powered language processing. This textbook offers an in-depth understanding of how neural models, transformers, and Large Language Models (LLMs) transform the way machines understand, generate, and interact with human language. Covering key concepts such as neural networks, word embeddings, pretraining, fine-tuning, prompting, augmentation, and efficiency of LLMs, as well as advanced topics and applications, combines theory with hands-on examples and explanations. Whether you’re new to the field or looking to advance your expertise, this book serves as an essential resource for navigating the cutting-edge of AI-driven communication.

Iryna Gurevych

Full Professor, Technical University Darmstadt
Past President, Association for Computational Linguistics

This book has a top-down structure that introduces foundational NLP concepts and gradually transitions into the complex topic of LLMs. By building intuition from basic topics, such as tokenisation, embeddings, and self-attention and layering more complex techniques like fine-tuning and transfer learning, readers develop a balance of conceptual and practical grasp of the subject. I find this book easy to read and would recommend that new beginners in the field utilise it as a resource to drive their intellect and become experts in LLMs. I hope they find the detailed diagrams and practice exercises as helpful as I found and utilise this book to its full potential.

Pushpak Bhattacharyya

Professor, Indian Institute of Technology Bombay
Past President, Association for Computational Linguistics

Preface

Throughout history, humankind's most remarkable innovations have often emerged from a need or a pressing problem that requires a solution. In the late prehistoric era, the challenge of transporting heavy goods more efficiently led to the invention of the wheel—a breakthrough that was later complemented by the steam engine for trains and the revolutionary invention of the aeroplane. Similarly, in the 20th century, the urgent need for fast and reliable communication across vast distances spurred the invention of the modern-day Internet, which in turn transformed global connectivity and information exchange. In recent times, this technological progress has accelerated, with the need for seamless communication and enhanced human experiences, ultimately leading to the development of Generative Artificial Intelligence (GenAI). With the ability to generate human-like interactions, realistic content, and automated creative processes, GenAI is arguably the most cogent innovation of the 21st century, which is steadily redefining communication in all aspects of human life.

Currently, the most noteworthy and widely applied contribution of GenAI happens to be the development of Large Language Models (LLMs). State-of-the-art general-purpose LLMs have proliferated in our daily lives at a rapid pace and play an integral role in carrying out both basic and highly complex tasks. The same models that can quickly draft content, answer frequently asked questions, and offer personalised recommendations based on the user's input are also capable of sifting through vast amounts of text to identify key patterns, generate scientific hypotheses, and assist professionals in diverse fields, such as medicine, technology, mental healthcare, and several others. The versatility of LLMs and their sophistication have led to their transformative impact on a global scale. Although they emerged from our need for smoother communication, LLMs now show incredible promise in expanding our knowledge across multiple fields and disciplines. As they stand today, LLMs may very well be the flagbearers of a potential technological revolution that is centred around GenAI.

Then, it is especially crucial to understand LLMs by assessing their known capacities and exploring what remains unknown. At present, we are witnessing what could be the very beginning of an AI-driven technological revolution. Experts from diverse fields have both applauded and warned against the ever-increasing applications of LLMs in a modern society. As powerful as they are, LLMs can pose threats to human safety and privacy if they are not understood and implemented conscientiously. At present, despite their immense popularity, LLMs are gravely misunderstood, and there is a lack of communication between AI developers and end-users to encourage the responsible development and usage of LLM-based tools. So, once again, the need

of the hour has led to the development of this book as a foundational and introductory guide to LLMs.

While writing the chapters of my book, I took a bottom-up approach, starting from basic yet essential topics like morphology and syntax and gradually progressing to more advanced subjects, such as retrieval-augmented generation and efficient fine-tuning of LLMs. This approach makes the book ideal for any novice enthusiast to deeply understand the topics and develop their perspective on handling LLMs. In recent years, there have been tremendous advancements in language modelling, requiring a strong understanding of theoretical and mathematical concepts to grasp the subject. However, presenting too much theory to a reader might cause boredom. Hence, I focused on exploring in-depth use cases of LLMs to keep the content engaging and break up any monotony. These use cases demonstrate LLMs' utility in areas like reasoning, model editing, and faithful evaluation. In the future, I believe AI safety will be the biggest ethical and technical problem of Natural Language Processing (NLP). It is important to be mindful of how this technology is used. Therefore, I designed chapters to implore the reader to think about such questions and focus on the ethical dilemmas LLMs might also face in the future.

BOOK OUTLINE

This book results from the course named '*Large Language Models: Introduction and Recent Advances*' that I started offering at the Indian Institute of Technology Delhi in the Monsoon Semester, 2024 (<https://lcs2-iitd.github.io/ELL881-AIL821-2401>). The course was mainly designed for senior undergraduate and graduate students as an advanced graduate-level course. There are fifteen chapters in the book, which can fall into four broad categories – (i) Foundations of Language Modelling, (ii) Core Concepts of LLMs, (iii) Efficient Utilisation of LLMs, and (iv) Applications of LLMs.

- 1. Foundations of Language Modelling:** Chapter 2 introduces the readers to basic concepts of NLP, neural networks, and evaluation metrics. These concepts are the building blocks of language modelling and are required to understand the rudimentary structure of language models. Chapter 3 builds upon these fundamentals and discusses word embeddings.
- 2. Core Concepts of Large Language Models:** Starting with Chapters 4 to 7, we brick by brick, build upon how LLMs came about by discussing early to current approaches. Chapter 4 begins with primitive n -gram models, laying the groundwork for understanding the evolution of deep learning-based models, including CNNs and RNNs, in Chapter 5. Chapter 6 introduces the self-attention mechanism, which forms the backbone of transformer models. Finally, in Chapter 7, we delve into the most widely adopted modelling techniques today: encoder-decoder architectures, autoencoders, and autoregressive models.
- 3. Efficient Utilisation of Large Language Models:** The next four chapters of the book deal with the best approaches on how to utilise the pretrained transformer models we learned about in Chapter 7. These topics can be broken down into fine-tuning mechanisms (Chapters 8 and 10) and prompting

strategies (Chapters 9 and 11). Here, the key focus is to develop the reader's understanding of how the models are currently being used in industry and help them develop their intellect in the field.

- 4. Applications of Large Language Models:** The last four chapters of the book deal with the use cases and limitations of LLMs that are the current focus of research in both industry and academia. We focus on how LLMs, which are generally trained on English data points and turn out to be biased can be expanded to deal with multilingual scenarios (Chapter 12) and be de-biased for social harmony (Chapter 13). We then move towards advanced, in Chapter 14, use cases of model editing and long context generations, these topics give the reader a clue of what must be done in the future to advance the LLMs technology and make them more apt to deal with our world. Our final chapter summarises the current state-of-the-art LLMs being used in research and development.

PREREQUISITES AND CONTENTS

Most of the chapters are self-explanatory and independent of any concepts outside the book. However, I encourage readers to take introductory courses in data structure and algorithms, linear algebra, calculus, and probability. Moreover, proficiency in Python programming is essential to fully take the benefit of this book. A few chapters require advanced skills. In a few sections of Chapters 11 and 12 and most of Chapter 14, readers are encouraged to learn the fundamentals of Torch programming and API calls to understand these sections. I encourage the reader to have a formal knowledge of deep learning and machine learning theory to fully enjoy the book in depth. I also expect the reader to be familiar with basic computer science terminologies and expect them to be familiar with how the LLMs superficially interact with humans.

This book is targeted towards senior undergraduates, graduates, and researchers. The book had a kept focus on research topics relevant to industry and academia, often referring to ideas, surveys, and tutorials published in natural language processing and machine learning-related venues like ACL, EMNLP, NAACL, NeurIPS, ICML, ICLR, etc.

BOOK LAYOUT

Each chapter accompanies a separate section, called 'additional information' containing—*reading materials* (scientific papers, blogs, websites, etc.) that are relevant to explore further details of the concepts covered in the chapter, *codes and datasets* that point to the source codes (released by the authors of the algorithms presented in the chapter) and publicly available datasets related to the chapter. The readers are highly encouraged to download the codes, run them on the datasets, and analyse the results. They may also try to reproduce the original publication results to understand the code's nitty-gritty, such as parameter selection, parameter tuning, and hyperparameter optimisation.

Advanced or difficult sections in different chapters are indicated with the ★ symbol. Beginners and novice readers may skip these sections. Exercises under each chapter are mostly divided into three parts: objective-type questions, short-answer questions, and long-answer questions.

OMISSIONS

This book is not comprehensive in covering all the topics in NLP and LLMs. In general, the area of LLMs is snowballing. Since this book's motivation is to present a foundation of LLMs, several topics are kept aside, which I deem to be outside the scope of a regular academic course. However, I mention a few of such essential topics in the additional material section of every chapter.

Omissions in every chapter are deliberate. For instance, there are plenty of studies on LLM safety, a few of which are discussed in Chapter 13. Several methods of model editing have not been presented in Chapter 14, which are relevant in advancing the field. Similarly, the architecture of LLMs is ever evolving with countless changes being proposed weekly, but I focus on the rudimentary structure of the models in Chapter 6 and briefly discuss alternate formulations of transformers that can be expanded on the reader with study in advance topics. I deliberately skip topics like a Mixture of Expert, State-Space Models, and key-query computation optimisation, which are advanced topics that may be relevant for the reader who wants to dive deeper into LLM architecture and computations. While I provide many applications of LLMs, this is not an exhaustive list. LLMs are being used on a multitude of tasks ranging from language modelling to robotics. A wise reader may study their sub-field of entrusting further after this book.

WEBSITE AND ONLINE MATERIALS

This book accompanies a website at <https://tanmoychak.com/llmbook/>. It accommodates additional lecture materials, such as lecture slides, lecture videos (<https://shorturl.at/4NwGE>), project ideas, reading materials, and recent research papers under every chapter. The website is being updated regularly with new materials. Readers may contact at book.largelanguagemodels@gmail.com for any queries or suggestions.

Acknowledgement

This book is far from the product of my efforts alone. It is the result of the collective dedication and support of so many, and I am deeply grateful to each and every one of them.

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I am deeply grateful to Prof. Timothy Baldwin for graciously writing the foreword to this book. His work has been a great source of inspiration throughout my journey in NLP. I still recall reading his seminal paper, ‘*Multiword Expressions: A Pain in the Neck for NLP*’, which ignited my passion and set me on the path to a career in NLP. I would also like to thank Prof. Iryna Gurevych and Prof. Pushpak Bhattacharyya for their kind endorsements of this book, as well as for the inspiring work they have contributed to the field—many of which have been referenced within this book.

To my parents, Ranjit Chakraborty and Anuradha Chakraborty, your endless love, belief in me, and unwavering encouragement have been my guiding light. You have shaped me in more ways than words can express.

A special thanks to my wife, Souravi, for your steadfast support, patience, and love through the long nights and challenging moments. Your belief in me has kept me going.

And finally, to my little Swastik—your laughter, joy, and boundless energy have been my greatest source of strength and inspiration. Even though you’re too young to understand, your presence has given me purpose every step of the way. This book carries a piece of the love and dedication I feel for you each day.

I sincerely owe my thanks to the team members of Wiley India and especially to Mr. Praveen Settigere and Mr. Vikash Kumar for helping me in publishing of this book.

Foreword

Like them or loath them, since they burst into the public consciousness in late 2022 with the release of ChatGPT, Large Language Models (LLMs) have quickly established themselves as an enabling technology for tasks ranging from document authoring, to information gathering, decision support, and document translation.

This book is a timely and accessible resource for understanding how modern-generation LLMs work. In particular, it walks the reader with a basic background in mathematics and computer science through the basics, covering the essential building blocks and milestone models that are used heavily in real-world applications, all the way to important ethical and social considerations in LLM deployment.

Starting with foundational principles in natural language processing, the book guides the reader through the basics of neural network models, tracing the evolution from statistical methods to advanced neural approaches. The introduction to transformer-based architectures sets the stage for understanding how advanced neural language models are pretrained on huge amounts of textual data, enabling the emergence of models like GPT and BERT that form the backbone of LLMs.

The book then delves into the mechanics of training and fine-tuning these models. From leveraging massive datasets to the nuances of pretraining encoder, decoder, and encoder-decoder models, along with the different stages of popular model training recipes, breaking the technical complexity down into accessible segments. Additionally, the reader is given valuable insights into evaluation techniques, highlighting the importance of metrics in refining and benchmarking model performance.

Beyond the technical mechanics, the book addresses the societal implications and responsibilities that tool developers should be aware of in LLM development. Critical issues, such as misinformation, social biases, and ethical considerations are scrutinised, highlighting the need to ensure that model deployment is aligned with human values and social principles as LLMs grow in capability.

Tanmoy Chakraborty has done a commendable job drawing together a remarkable breadth of material in this volume. He offers a coherent, engaging, and insightful perspective that serves not just as a guide for beginners but also as a one-stop-shop reference for researchers and practitioners. Whether you are interested in LLMs out of academic curiosity, for practical applications, or to achieve societal impact, this book will be a valuable companion.

Timothy Baldwin

Professor and Provost, Mohamed bin Zayed University of Artificial Intelligence
Melbourne Laureate Professor, The University of Melbourne
Past President, The Association for Computational Linguistics

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